# Ahan Mondal

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## Education

**Kalinga Institute of Industrial Technology** — Bhubaneswar, Odisha
*Bachelor of Technology in Computer Science — CGPA: 8.86 (6 Semesters)* — *Aug. 2023 – Present*

**Salt Lake School** — Kolkata, West Bengal
*Class 12 (ISC) – 89.5%(Best of 4)* — *Class 10 (ICSE) – 95.6%(Best of 5)* — *2008 – 2023*

## Skills

**Languages**: Java, Python, C, SQL, JavaScript, Shell Script
**Core Concepts**: DSA, OOP, DBMS, CN, Deep Learning, Graph Neural Networks, LangGraph, REST API Design, JWT Authentication
**Frameworks & Libraries**: PyTorch, PyTorch Geometric, Pandas, NumPy, Scikit-learn, CatBoost, Matplotlib, Streamlit, Node.js, Express.js
**Databases**: SQLite, SQL(Oracle), MariaDB
**Tools**: Power BI, Excel, Git, Linux, Vercel, Vite, Vim, VS Code, Jupyter Notebook, AWS S3

## Experience

**Coal India Limited (CIL) — System Department** — Kolkata, West Bengal
*Software Engineering Intern* — *May 2026 – Jun. 2026*

- Built a full-stack TPSA Agreement Management System for CIL (300+ mines) using **Node.js**, **Express.js**, **React**, and **MariaDB** with JWT authentication and role-based access control.
- Engineered multi-party e-signature workflows, automated PDF generation from DOCX templates via headless **LibreOffice**, and secure file serving through expiring signed token URLs.
- Deployed **AWS S3** cloud storage, integrated external REST APIs for OTP-based onboarding, and implemented append-only audit logging for enterprise data governance compliance.

## Projects

**MedPath AI [Live]** | *FastAPI, React 19, LangGraph, Gemini 2.5 Flash, Supabase, JWT* — May 2026

- Led chatbot development for a 3-person team using a 4-node **LangGraph** state graph with **Gemini 2.5 Flash**, delivering emergency bypass detection across 500+ hospital records.
- Worked with the team on a multi-factor hospital scoring engine using Pandas and Haversine distance, with comorbidity-based cost multipliers for diabetes (+12%) and cardiac history (+18%); enabling Supabase-backed loan application and status tracking with 3-second polling updates.
- Coordinated with backend engineer to connect a loan pre-qualification pipeline with 4 underwriting checks and PDF data extraction, reducing manual eligibility review to under 1 minute with instant GREEN/YELLOW/RED decisions

**Vehicle Insurance Fraud Detection** | *Python, PyTorch, CatBoost, PyTorch Geometric, Scikit-learn* — Mar. 2026

- Architected a multi-modal fusion model as part of a 4-person team, combining 16-dimensional CatBoost tabular embeddings, 16-dimensional LSTM temporal embeddings, and 32-dimensional GCN graph embeddings into a 128-64-1 DNN classifier.
- Owned data preprocessing and LSTM temporal branch in a collaborative pipeline; engineered stratified splitting, weighted loss functions, and focal-loss experimentation achieving ROC-AUC 0.8291 and PR-AUC 0.2259.
- Constructed a threshold-tuning workflow over 91 candidates from 0.05 to 0.95, selecting a precision-first threshold of 0.54 at 0.80 recall and a cost-optimal threshold of 0.39 with FP/FN cost 1:25 and business cost 1489.0.

## Certifications & Achievements

**Cisco CCNA** | *Cisco Networking Academy* — Jun. 2026

- Introduction to Networks
- Switching, Routing & Wireless Essentials
- Enterprise Networking, Security & Automation

**AlgoUniversity Graphs Contest** | *Rank 10 (College), Scholarship Consideration*